$$p(r|d) = \frac{p(d|r)\,p(r)}{p(d)} \qquad p(x_1, x_2) = p(x_1)\cdot p(x_2)$$

if $p(r|d,q) > p(\neg r|d,q)$

$$\frac{p(r|d)}{p(\neg r|d)} = \frac{p(d|r)\cdot p(r)/\cancel{p(d)}}{p(d|\neg r)\cdot p(\neg r)/\cancel{p(d)}} = \frac{p(d|r)\cdot p(r)}{p(d|\neg r)\cdot p(\neg r)} \overset{?}{>} 1$$

$$= \frac{\prod p(x_i|r)\cdot p(r)}{\prod p(x_i|\neg r)\cdot p(\neg r)}$$

$$p(d|r) = p(x_1, x_2 \cdots x_n|r)$$
$$= p(x_1|r)\cdot p(x_2|r)\cdots p(x_n|r)$$
$$= \prod p(x_i|r)$$

$$p(x_i|r) = \begin{cases} a_i & x_i = 1 \\ 1-a_i & x_i = 0 \end{cases} \qquad p(d|r) = \prod_{i=1\cdots n}\left[a_i^{x_i}(1-a_i)^{1-x_i}\right]$$

$$p(d|\neg r) = \prod\left[b_i^{x_i}(1-b_i)^{1-x_i}\right]$$

$$c_i = \log\frac{a_i(1-b_i)}{b_i(1-a_i)}$$

$$C = \sum \log\frac{1-a_i}{1-b_i}$$

$$= \prod \frac{\left[a_i^{x_i}(1-a_i)^{1-x_i}\right]}{\left[b_i^{x_i}(1-b_i)^{1-x_i}\right]} > 1$$

$$= \prod \frac{a_i^{x_i}(1-a_i)(1-b_i)^{x_i}}{b_i^{x_i}(1-b_i)(1-a_i)^{x_i}} \overset{\log}{=} \prod\left[\frac{a_i(1-b_i)}{b_i(1-a_i)}\right]^{x_i}\cdot\prod\frac{1-a_i}{1-b_i} > 1$$

$$= \sum x_i \underbrace{\log\frac{a_i(1-b_i)}{b_i(1-a_i)}}_{x_i} + \sum\log\frac{1-a_i}{1-b_i} > 0 \qquad = \sum c_i x_i + C$$

$$= R(d,q)$$